Introduction to machine learning competition project

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1. Introduction

1.1. Task Description

The goal of this competition was to build an image retrieval system capable of bridging a significant real-to-synthetic domain shift. We were asked to take a query image a real photograph of a celebrity and identify the most similar matches from a gallery set. The difficulty stemmed from the fact that the gallery was composed entirely of synthetic versions of those celebrities, including 3D renders, AI-generated images, and stylized illustrations.
Unlike standard classification, this task required precise identity matching across visually different domains. The dataset provided a training split of about 5900 images (natural and synthetic) to help us understand these relationships. The final evaluation was conducted on a test set of 1456 real queries and 1480 synthetic gallery images. The main technical hurdle was that real photos contain natural details like skin texture and complex lighting that are absent or distorted in synthetic renders. Managing this within a strict two-hour window forced us to prioritize models that could handle these stylistic discrepancies efficiently.

1.2. Overview of Approaches

To address this task, we experimented with three main strategies. First, we used CLIP in a zero-shot setting, hoping its massive pre-training on diverse data would allow it to recognize identity while ignoring stylistic changes. Second, we implemented Siamese Networks using a ResNet18 backbone to explicitly teach the model that a real photo and a synthetic render of the same person belong together. Finally, we used standard ResNet architectures (both pre-trained and fine-tuned) as a baseline to see if traditional convolutional features could hold up under the domain shift.

1.3. Summary of Results

The competition server showed that CLIP Zero-Shot was the most effective approach by a wide margin. It achieved a score of 345.57, significantly outperforming our other attempts. While the Siamese and ResNet models provided a baseline, they struggled to generalize across the real-to-synthetic gap in such a short amount of time. CLIP's ability to grasp the ”concept” of the person's face proved much more robust than the texture-based features of the other models.

2. Models Considered

In this section, we describe the models evaluated for the instance-level image retrieval task, outlining their theoretical foundations and relevant literature. The selected approaches span classification-pretrained convolutional networks, metric-learning frameworks, self-supervised contrastive methods, and large-scale multimodal models. These paradigms represent different strategies for structuring embedding spaces, allowing us to compare how their training objectives influence retrieval performance under real-to-synthetic domain shift. Although several architectures were explored during preparation, only a subset was fully deployed during the competition due to the strict two-hour time constraint. The remaining models are described to contextualize the design space considered rather than to report fully optimized results.

2.1. CLIP (Contrastive Language-Image Pretraining)

Description: CLIP (Radford et al., 2021) was our most successful model. It uses a ViT-L/14 backbone trained on 400 million image-text pairs. We chose it because its contrastive pre-training allows it to learn high-level identity features. For our implementation, we used the model in a zero-shot configuration, extracting 768 dimensional embeddings which were L2 normalized to compute cosine similarity. The contrastive objective helps organize the embedding space so that semantically related images remain close even when low-level visual details differ. This allows CLIP to rely more on global identity structure rather than purely textural patterns, making it more robust to stylistic discrepancies compared to classification-pretrained CNNs.

Reference: Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., ... & Sutskever, I.

(2021). Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*.

2.2. Siamese Networks

Description: Siamese Networks (Koch et al., 2015) consist of two identical subnetworks sharing parameters and trained using contrastive loss. We implemented a Siamese Network using a ResNet18 backbone. We optimized the architecture using a Contrastive Loss function, which acts as a distance-based regularizer. By enforcing a margin m, the model attempts to map the intra-class variance below the inter-class separation a fundamental requirement for effective identity retrieval in unseen domains. The objective was to minimize the distance between positive pairs (same identity) and maximize it for negative pairs. We used a margin $m = 1.0$, the Adam optimizer with a learning rate of 10^{-4}, and a batch size of 32 for 5 epochs. Due to the two-hour limit, we could only train on a limited number of pairs, which significantly impacted the model's ability to reach convergence.

Reference: Koch, G., Zemel, R., & Salakhutdinov, R. (2015). Siamese neural networks for one-shot image recognition. In *ICML Workshop*.

2.3. ResNet (Pretrained & Fine-Tuned)

Description: We used ResNet18 and ResNet50 pretrained on ImageNet. For the fine-tuned version, we retrained the model on the 5900 training images using Cross-Entropy Loss for 10 epochs with a learning rate of 0.00001. We then removed the final classification head to use the remaining layers as a feature extractor. However, optimizing cross-entropy for identity classification does not explicitly enforce geometric separation in embedding space, which is critical for retrieval.

Reference: He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

2.4. Vision Transformers (HuggingFace ViT)

In addition to CLIP and ResNet, we deployed a Vision Transformer (ViT) from the HuggingFace library as an alternative feature extractor. Specifically, we utilized the vit-base-patch16-224 model pretrained on ImageNet-21k. Unlike standard CNNs, which rely on local convolutional filters, ViT uses self-attention mechanisms to capture global dependencies across the entire image. This architectural difference allowed the model to focus on the overall facial structure and geometry rather than being distracted by the local texture discrepancies caused by the real-to-synthetic domain shift. We extracted the embeddings from the classification token, resulting in a 768 dimensional vector that achieved a competitive score of 43.37.

Reference: Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*.

2.5. Preliminary Explorations (Not Deployed)

During the initial preparation phase, we considered several other metric-learning and self-supervised architectures, such as ArcFace, Triplet Networks, and SimCLR. While these models are theoretically well-suited for instance-level retrieval and identity matching, they require extensive "hard-negative mining" and long convergence times to effectively structure the embedding space. Given the strict two-hour competition limit, we decided not to deploy them as we could not ensure stable training or proper hyperparameter tuning within the window. Instead, we prioritized the robust zero-shot capabilities of CLIP and the more straightforward adaptation of the Siamese and ResNet pipelines.

3. Evaluation

All models were evaluated under the same retrieval protocol.

Performance was measured using Top-5 accuracy, defined as the proportion of query images for which the correct identity appears among the top-5 retrieved gallery images. With 1456 queries and k=5, the maximum number of possible correct matches is 7280. This provides a reference scale for interpreting the absolute scores returned by the competition server.
For metric-learning approaches, lightweight training configurations were adopted to ensure stable execution within the two-hour competition window. These included 5-10 training epochs, batch sizes of 16 or 32, learning rates of 10^{-4} or 10^{-5}, and an embedding dimension of 256. Approximately 2,000 training pairs were used when applicable.

3.1. Quantitative Results

The values reported in Table 1 correspond to the official Top-5 scores provided by the competition server. These scores reflect the total number of successful retrievals across all queries, weighted according to ranking performance.

Table 1 reports the Top-5 retrieval accuracy obtained by the evaluated models.

CLIP in a zero-shot configuration achieved the highest retrieval accuracy by a substantial margin. The performance

Table 1. Top-5 accuracy in the image retrieval task. ND: not deployed during the competition.

Model	Top-5 Accuracy
CLIP (Zero-Shot)	345.57
Siamese Networks	46.52
HuggingFace ViT	43.37
ResNet18	30.72
ResNet50 (Pretrained)	30.58
FAISS	22.75
ResNet50 (Fine-Tuned)	ND
CLIP (Fine-Tuned)	ND
Triplet Network	ND
ArcFace	ND
SimCLR	ND
Timm	ND

gap between CLIP and the next-best baseline (Siamese Networks) indicates that the quality of the representation and the pretraining objective play a decisive role in this retrieval setting.

CNNs pretrained in classification (ResNet variants) obtained considerably lower accuracy, suggesting that category-level training objectives are less suited for identity-level retrieval under a real-to-synthetic domain shift. In absolute terms, CLIP retrieves more than 300 correct matches, while Siamese remains below 50 and ResNet variants remain around 30. This gap indicates a structural difference in embedding quality rather than a marginal improvement.

3.2. Qualitative Observations

Qualitative inspection of the retrieved results further supports the quantitative findings. CLIP embeddings were generally robust to stylistic discrepancies between real query images and synthetic gallery images. Correct matches were often retrieved despite variations in pose, illumination, or rendering style.

In contrast, CNNs pretrained in classification frequently retrieved visually similar but identity-mismatched synthetic images, indicating sensitivity to low-level visual features rather than identity-level structure.

4. Final Discussion

The competition highlighted the importance of picking the right model for the specific domain. In a real-to-synthetic task, traditional CNNs like ResNet are at a disadvantage because they are too sensitive to texture changes. On the other hand, foundation models like CLIP are much more robust because they prioritize identity over style.

The two-hour time limit was a deciding factor. This demonstrated that strong pretrained representations can outperform partially optimized specialized architectures under strict time constraints. For future work, we would focus on using CLIP as a base and applying a very light fine-tuning (like LoRA) to adapt it to the specific celebrity faces in the dataset, while maintaining its robustness to style changes.
Overall, this project made it clear that the choice of training objective can matter more than architectural complexity when dealing with cross-domain retrieval.
Future work could explore controlled fine-tuning of CLIP on identity-level supervision, more systematic hyper-parameter optimization for metric-learning methods, or hybrid pipelines combining contrastive pretraining with identity-specific adaptation.

Despite its success, CLIP occasionally struggled with identities featuring heavy occlusions or extreme lighting variations in the synthetic domain. This suggests that while semantic embeddings are robust to style, they still benefit from domain-specific structural alignment.

5. Workload Table

The following table summarizes the contributions of each team member. The Models row indicates the models developed by each member that were deployed during the competition. The Models ND (Not Deployed) row lists models implemented but not used in the final submission. The Report row reflects each member's contribution to the writing and revision of this document, while the Repository row specifies contributions to the development and organization of the project's GitHub repository.

Table 2. Models' accuracy in the image retrieval task. ND: not deployed in the competition.

Member	Contributions
Tommaso Grotto	Models: CLIP, HuggingFace ViT, ResNet50 Models ND: Triplet Network, ArcFace, SimCLR Report: 20% Repository: 30%
Stefano Murtas	Models: ResNet18, FAISS, Siamese Networks Models ND: Timm Report: 80% Repository: 70%

The code used for this project is available at: https://github.com/smurtas/ML-STF.git